

## Assignment 19.2 Ai Assisted Coding

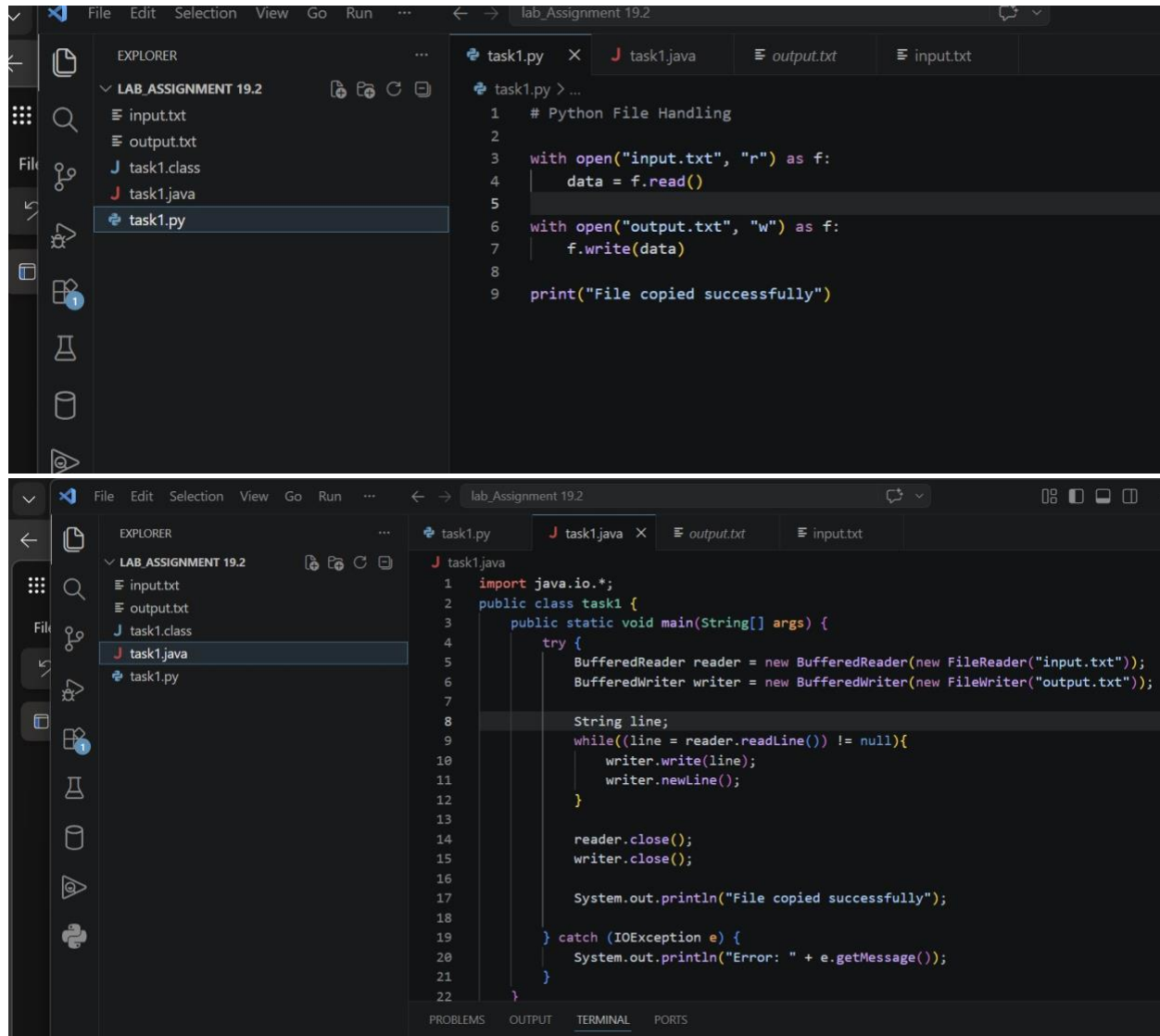
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Task 1: Python to Java Conversion (File Handling) Prompt:

Convert a Python file handling program into Java.

Code:



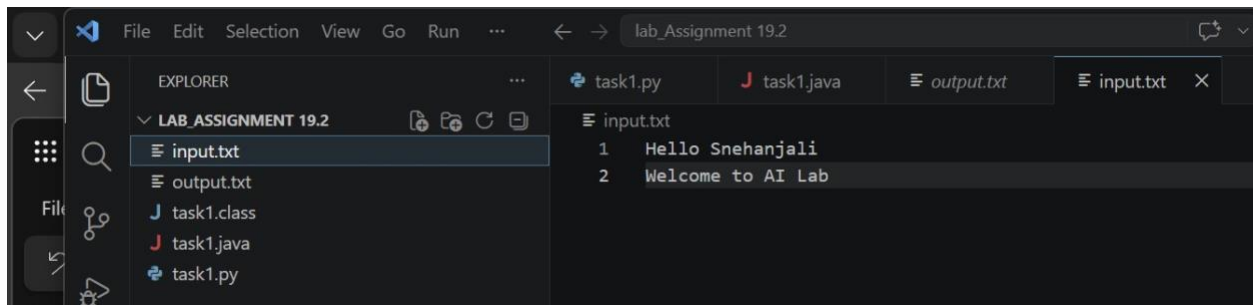
The image consists of two screenshots of an IDE, likely Visual Studio Code, showing the conversion of a Python file handling program into Java.

The top screenshot shows the Python code in `task1.py`. The code reads from `input.txt` and writes to `output.txt`.

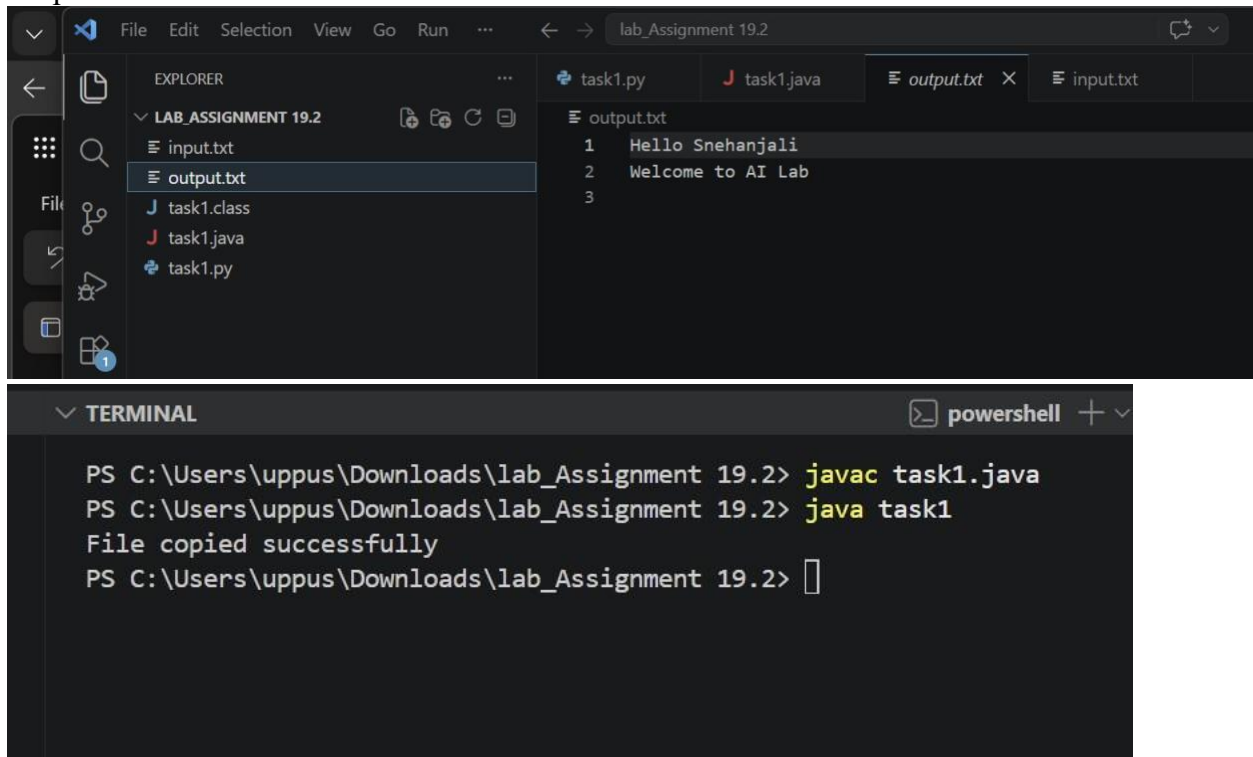
```
1 # Python File Handling
2
3 with open("input.txt", "r") as f:
4     data = f.read()
5
6 with open("output.txt", "w") as f:
7     f.write(data)
8
9 print("File copied successfully")
```

The bottom screenshot shows the Java code in `task1.java`. The code uses `BufferedReader` and `BufferedWriter` to read from `input.txt` and write to `output.txt`.

```
1 import java.io.*;
2 public class task1 {
3     public static void main(String[] args) {
4         try {
5             BufferedReader reader = new BufferedReader(new FileReader("input.txt"));
6             BufferedWriter writer = new BufferedWriter(new FileWriter("output.txt"));
7
8             String line;
9             while((line = reader.readLine()) != null){
10                 writer.write(line);
11                 writer.newLine();
12             }
13
14             reader.close();
15             writer.close();
16
17             System.out.println("File copied successfully");
18
19         } catch (IOException e) {
20             System.out.println("Error: " + e.getMessage());
21         }
22     }
23 }
```



Output:



Explanation:

The Python program reads data from a file and writes it to another file using open() function.

The Java program uses BufferedReader and BufferedWriter for file handling.

Exception handling is required in Java.

Both programs perform the same operation and produce identical output.

Task2: C++ to Python Conversion (Sorting)

Prompt:

Convert a C++ sorting program into Python.

Code:

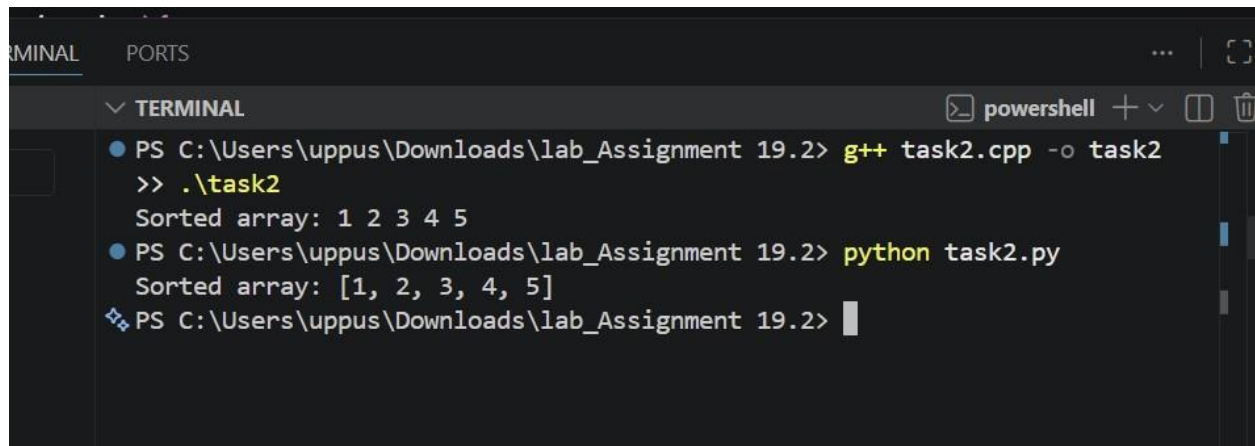
The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying files for 'LAB\_ASSIGNMENT 19.2': input.txt, output.txt, task1.class, task1.java, task1.py, task2.cpp (selected), task2.exe, and task2.py. The main editor window shows the code for task2.cpp. The code implements a bubble sort algorithm in C++.

```
task2.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[5] = {5, 2, 4, 1, 3};
6
7      for(int i=0;i<5;i++){
8          for(int j=0;j<5-i-1;j++){
9              if(arr[j] > arr[j+1]){
10                 swap(arr[j], arr[j+1]);
11             }
12         }
13     }
14
15     cout << "Sorted array: ";
16     for(int i=0;i<5;i++)
17         cout << arr[i] << " ";
18
19     return 0;
20 }
```

The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying files for 'LAB\_ASSIGNMENT 19.2': input.txt, output.txt, task1.class, task1.java, task1.py, task2.cpp, task2.exe, and task2.py (selected). The main editor window shows the code for task2.py. The code implements a bubble sort algorithm in Python.

```
task2.py > ...
1  arr = [5, 2, 4, 1, 3]
2
3  for i in range(len(arr)):
4      for j in range(len(arr) - i - 1):
5          if arr[j] > arr[j + 1]:
6              temp = arr[j]
7              arr[j] = arr[j + 1]
8              arr[j + 1] = temp
9
10 print("Sorted array:", arr)
```

Output:



A terminal window titled 'TERMINAL' with a 'powershell' icon. It shows three commands and their outputs in a Windows PowerShell environment. The first command compiles a C++ file, the second runs it, and the third runs a Python script. Both programs output a sorted array of the numbers 1 through 5.

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task2.cpp -o task2
>> .\task2
Sorted array: 1 2 3 4 5
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task2.py
Sorted array: [1, 2, 3, 4, 5]
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The C++ program uses Bubble Sort to sort an array.

-->The Python program follows the same logic.

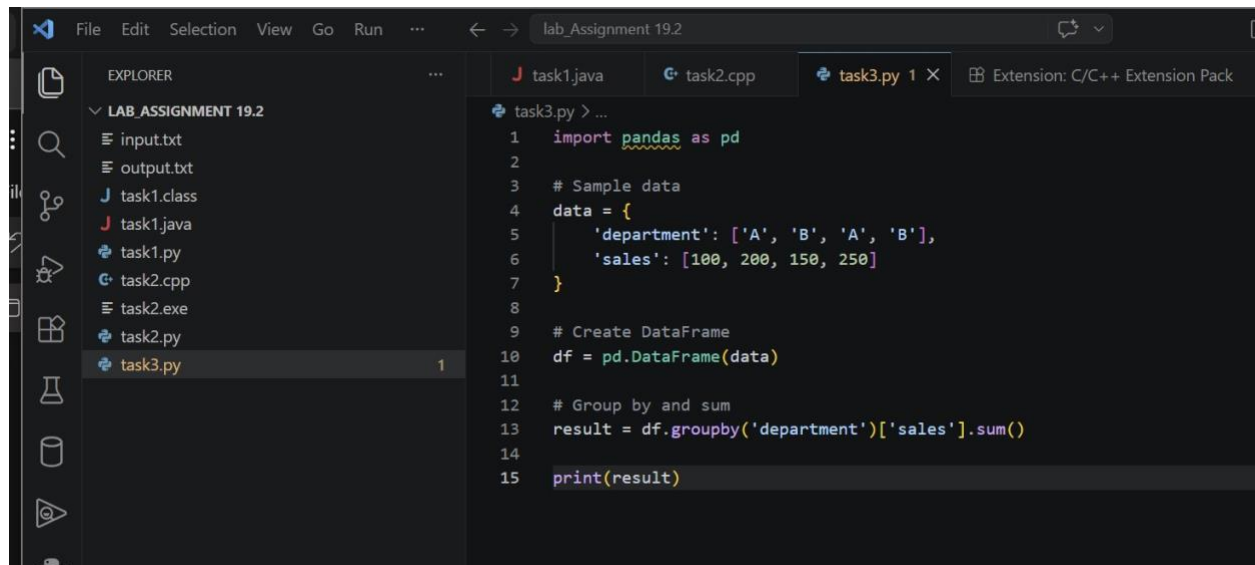
-->Both programs produce the same sorted result.

### Task 3: SQL to Pandas Conversion

Prompt:

Convert a SQL query with GROUP BY into Pandas.

Code:



A screenshot of the Visual Studio Code editor. The Explorer sidebar on the left shows a project named 'LAB\_ASSIGNMENT 19.2' with files including 'input.txt', 'output.txt', 'task1.class', 'task1.java', 'task1.py', 'task2.cpp', 'task2.exe', 'task2.py', and 'task3.py'. The main editor window is open to 'task3.py', which contains Python code using pandas to convert a SQL GROUP BY query. The code imports pandas, defines a sample dataset with departments and sales, creates a DataFrame, and then groups the data by department to sum the sales.

```
1 import pandas as pd
2
3 # Sample data
4 data = {
5     'department': ['A', 'B', 'A', 'B'],
6     'sales': [100, 200, 150, 250]
7 }
8
9 # Create DataFrame
10 df = pd.DataFrame(data)
11
12 # Group by and sum
13 result = df.groupby('department')['sales'].sum()
14
15 print(result)
```

Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task3.py
department
A    250
B    450
Name: sales, dtype: int64
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The SQL query groups data by department and calculates total sales.

-->The Pandas code uses `groupby()` and `sum()` to achieve the same result.

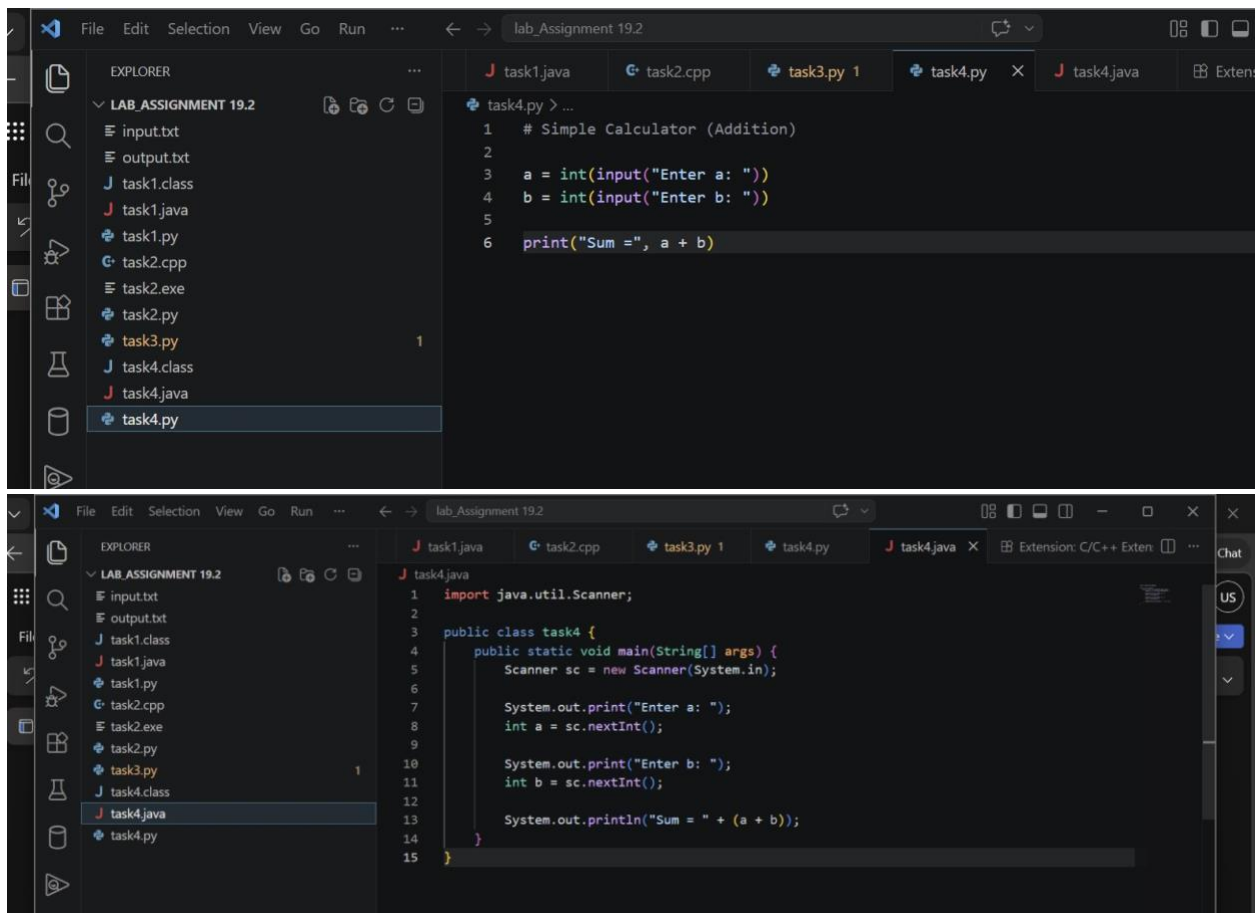
-->Both produce identical outputs.

#### Task 4: Real-Time Code Conversion (Python to Java)

Prompt:

Convert a Python calculator program into Java.

Code:



Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task4.py
Enter a: 3
Enter b: 5
Sum = 8
```

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> javac task4.java
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> java task4
Enter a: 3
Enter b: 5
Sum = 8
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The Python program performs addition using simple input and print statements.

-->The Java program uses Scanner for input and follows class-based structure.

-->Both programs perform the same operation and produce identical output.

## Task5: Python to C++ Conversion (Even/Odd)

### Prompt:

Convert a Python program to C++ to check even or odd number.

### Code:

The image consists of two screenshots of a code editor, likely Visual Studio Code, showing the conversion of a Python program to C++.

**Top Screenshot:** The editor shows a file named `task5.py` open. The Explorer panel on the left lists files for `LAB_ASSIGNMENT 19.2`, including `input.txt`, `output.txt`, `task1.class`, `task1.java`, `task1.py`, `task2.cpp`, `task2.exe`, `task2.py`, `task3.py` (highlighted), `task4.class`, `task4.java`, `task4.py`, `task5.cpp`, `task5.exe`, and `task5.py`. The main editor area shows the Python code for `task5.py`:

```
1 # Check Even or Odd
2
3 n = int(input("Enter number: "))
4
5 if n % 2 == 0:
6     print("Even")
7 else:
8     print("Odd")
```

**Bottom Screenshot:** The editor shows a file named `task5.cpp` open. The Explorer panel on the left is the same as the top screenshot, but `task5.cpp` is now highlighted. The main editor area shows the C++ code for `task5.cpp`:

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n;
6     cout << "Enter number: ";
7     cin >> n;
8
9     if(n % 2 == 0)
10         cout << "Even";
11     else
12         cout << "Odd";
13
14     return 0;
15 }
```

### Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task5.py
Enter number: 3
Odd
● PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task5.cpp -o task5
>> .\task5
Enter number: 3
○ Odd
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The Python program checks whether a number is even or odd using modulus operator.

-->The C++ program follows the same logic using if-else condition.

-->Both programs produce correct results for all test cases.